

QUANTUMSHIELD

Quantum-Safe Cryptographic Bill of Materials (CBOM) & Risk Assessment Report

1. Executive Summary & Project Context

Project Name:	QuantumShield Demo Project
Source Type:	DEMO
Report Date:	2026-09-10 17:12 UTC
Total Crypto Assets:	63

2. Cryptographic Risk & Priority Distribution

Metric	Count	Classification Note
Quantum Vulnerable Assets	15	RSA, ECC, ECDSA, ECDH, DH vulnerable to Shor's Algorithm
Classical Weaknesses	7	MD5, SHA-1, DES legacy algorithms
Monitor Items	14	AES-256 / Modern hashes (Monitor key length & margins)
P0 Immediate Planning	0	Critical business exposure + long data lifetime
P1 High Priority	22	High risk assets in active business paths

3. Discovered Cryptographic Inventory (CBOM)

Algorithm	Purpose	Risk Type	Score	Priority	File Location
OPENSSL	TLS	Unknown	52.5	P2	Dockerfile:4
X.509 CERTIFICATE	TLS	Unknown	52.5	P2	Dockerfile:5
CERTBOT	TLS	Unknown	52.5	P2	Dockerfile:6
CRYPTOJS	Unknown	Unknown	52.5	P2	package.json:6
JSONWEBTOKEN	Unknown	Unknown	52.5	P2	package.json:7
BOUNCY CASTLE	Unknown	Unknown	52.5	P2	pom.xml:8
BOUNCY CASTLE	Unknown	Unknown	52.5	P2	pom.xml:9
APACHE COMMONS CODEC	Unknown	Unknown	52.5	P2	pom.xml:13
APACHE COMMONS CODEC	Unknown	Unknown	52.5	P2	pom.xml:14
PYCA CRYPTOGRAPHY	Unknown	Unknown	52.5	P2	requirements.txt:1

PYCRYPTODOME	Unknown	Unknown	52.5	P2	requirements.txt:2
PYCRYPTO	Unknown	Unknown	52.5	P2	requirements.txt:2
PYJWT	Unknown	Unknown	52.5	P2	requirements.txt:3
UNKNOWN	Key Generation	Unknown	52.5	P2	java/CryptoService.java:3
UNKNOWN	Key Generation	Unknown	52.5	P2	java/CryptoService.java:4

4. Recommended Post-Quantum Alternatives (NIST FIPS Standardized)

Current Primitive	Purpose	Candidate NIST PQC Target	Migration Complexity
OPENSSL	TLS	Hybrid Post-Quantum TLS 1.3 (e.g. X25519 + ML-KEM-768)	Medium
X.509 CERTIFICATE	TLS	Hybrid Post-Quantum TLS 1.3 (e.g. X25519 + ML-KEM-768)	Medium
CERTBOT	TLS	Hybrid Post-Quantum TLS 1.3 (e.g. X25519 + ML-KEM-768)	Medium
CRYPTOJS	Unknown	Manual Cryptographic Review Required	Medium
JSONWEBTOKEN	Unknown	Manual Cryptographic Review Required	Medium
BOUNCY CASTLE	Unknown	Manual Cryptographic Review Required	Medium
BOUNCY CASTLE	Unknown	Manual Cryptographic Review Required	Medium
APACHE COMMONS CODEC	Unknown	Manual Cryptographic Review Required	Medium

5. Mosca Theorem Migration Roadmap

Planning Threat Horizon: 10 years

Assessment Rationale: URGENT: Required security window (12 years = 10 yrs data lifetime + 2 yrs migration) exceeds the configured planning threat horizon (10 years). Data protected today may be exposed to 'Harvest Now, Decrypt Later' quantum attacks.

***Disclaimer:** QuantumShield candidate migration guidance is provided for security planning and risk prioritizing purposes only. All cryptographic algorithm replacements must be validated in non-production environments before deployment.*